

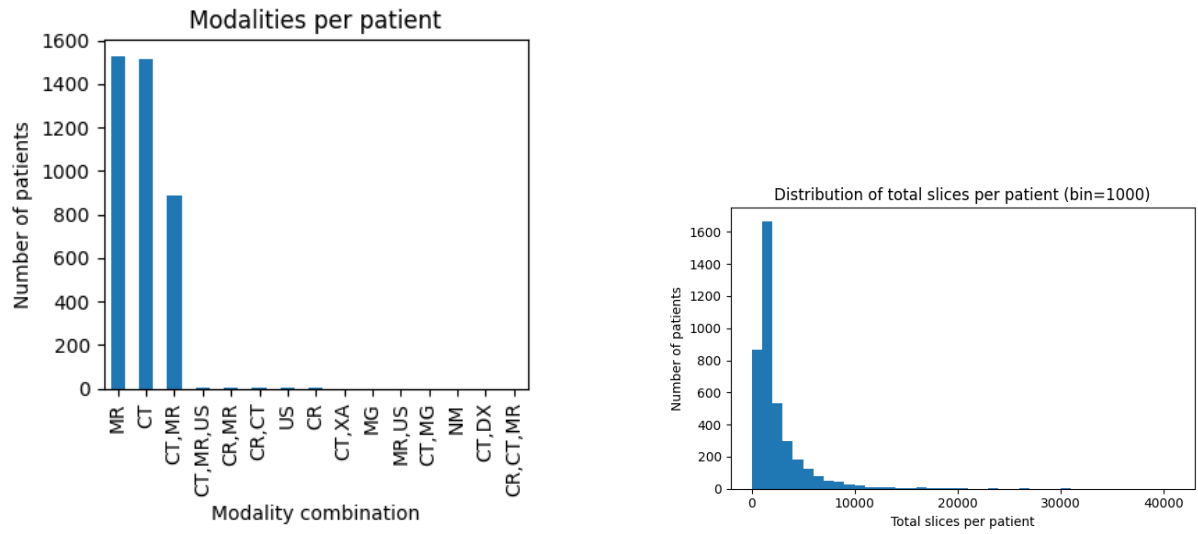
Pre-processing of image and its metadata

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1 Extraction

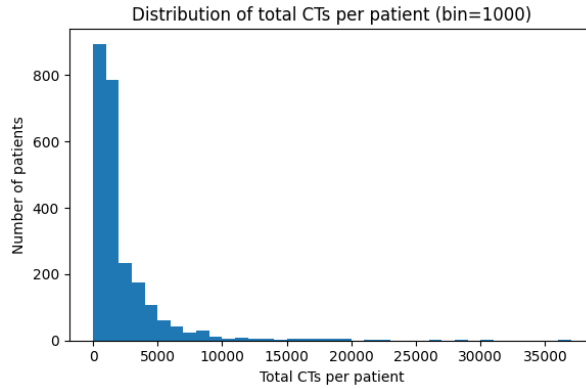
1.1 Extraction of valid patients

We conclude that there are total 4904 qualified AP/CP patients derived from the clinical dataset. After the initial extraction of medical images according to their "MRN", 3955 out of 4904 patients have at least one medical image. Here are the distribution of slices and modalities:



1.2 Extraction of CT only patients

Since our research only focus on CT analysis, we extract all patients with CT data, and there are 2417 patients left. Here are the distribution of CT:



2 Body Part Inference and Denoise

As shown in the distribution, most patients have thousands of CT slices. Therefore, it is necessary to develop an inference method that can precisely label the anatomical region of each slice (e.g., identifying pancreas-specific slices rather than broadly labeling them as abdomen). This enables the removal of slices that are clearly unrelated to the research objective. Guided by domain knowledge and empirical inspection of the metadata, we first extracted the following informative metadata fields:

Anatomical Metadata

- Body Part Examined
- Series Description
- Study Description
- Protocol Name
- Performed Protocol Code Sequence
- Scheduled Protocol Code Sequence

Geometry and Spatial Information

- Slice Position (Z)
- Image Orientation (Patient Coordinate System)
- Pixel Spacing
- Slice Thickness

Image Type and Classification

- Image Type

Acquisition Parameters

- Series Number
- Acquisition Number
- Reconstruction Kernel

2.1 Anatomical Region Inference

We labeled each imaging series with an anatomical region using a rule-based method that combines DICOM metadata and scan descriptions. More reliable signals were used first, and less strict rules were applied only when needed to limit misclassification.

First, series corresponding to non-diagnostic images (e.g., localizers, scout scans, or reformatted views such as MPR/MIP/VR) were identified via the **ImageType** field and excluded from downstream analysis. Next, when available, the **BodyPartExamined** tag was normalized to a predefined set of canonical anatomical categories (e.g., Abdomen, Abdomen+Pelvis, Pancreas, Chest, Spine) using a curated synonym mapping to account for common institutional and vendor-specific variations.

If **BodyPartExamined** was missing or uninformative, we inferred the anatomical region from free-text fields (**SeriesDescription**, **StudyDescription**, and **ProtocolName**). These fields were searched for conservative, domain-specific keyword patterns corresponding to pancreas-focused studies, multi-region protocols (e.g., chest–abdomen–pelvis), cardiovascular imaging, spine, head/brain, and extremity examinations. Priority was given to pancreas-related patterns, followed by multi-region and single-region detections. Ambiguous or unclassifiable series were labeled as unknown and excluded.

This hierarchical approach enables robust anatomical labeling across heterogeneous imaging protocols while maintaining conservative inclusion criteria for region-specific analyses. Geometric metadata were

not required for additional inference when `BodyPartExamined` was unavailable, as anatomical region labels could be reliably inferred for all slices.

Body Region	Count
Abdomen	3,436,564
Reformatted	1,188,239
Chest	384,479
Abdomen+Pelvis	60,711
Pancreas	52,401
CHABPE	29,416
Heart	19,852
Localizer	10,975
Head	8,765
Extremity	5,317
Aorta	5,061
Spine	4,021
Neck	3,922
Pelvis	2,313

2.2 Filter irrelevant slices

Since pancreatitis is localized to the pancreas and surrounding abdominal structures, imaging of distant anatomical regions is not expected to provide predictive information for disease progression and may dilute pancreas-specific signals.

For downstream analysis, we restricted the dataset to imaging series corresponding to anatomical regions of primary clinical interest. Specifically, we retained only series assigned to the following canonical body-region categories: `PANCREAS`, `ABDOMEN`, `ABDOMEN+PELVIS`, `CHABPE`, and `AORTA`. Series labeled outside this set were excluded from further analysis.

As a result, 2,300 out of 2,417 patients retained at least one qualifying imaging series. This modest reduction in sample size is acceptable given the favorable trade-off between data quantity and noise reduction. Moreover, the number of patients with extremely large numbers of CT slices decreased substantially, as illustrated in the chart below.

